

A parameter-free, pre-registered test that the galaxy acceleration scale tracks the dark-energy density:

$$a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$$

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Abstract

The galaxy acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ that sets the onset of the mass-discrepancy/MOND regime coincides numerically with $c\sqrt{\rho_\Lambda}$ — a fact noted by Limbach, Psaltis & Özel (2008). In the de Sitter–Unruh modified-inertia framework this scale is sourced by the cosmological horizon, $a_0 = cH_\Lambda/Z$; promoting it adiabatically as the horizon evolves ties the coupling, under the adiabatic-horizon ansatz (itself posited, not derived), to the dark-energy density and yields a parameter-free redshift law, $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$. We study the ratio $R \equiv a_0(z)/a_0(0)$, which is independent of the framework’s internal normalization Z and of the present-day value of a_0 (both cancel), so that it depends only on the measured dark-energy equation of state $w(z)$. This makes the framework falsifiable: R tracks $\sqrt{\rho_{\text{DE}}(z)}$; under the DESI-preferred ($w_0 > -1$, $w_a < 0$) histories it declines at high redshift, it reduces to $R = 1$ exactly if $w = -1$ (constant- a_0 MOND), and a measured rise registers as a strained verdict in the frozen protocol. We confront the prediction non-circularly across two scales — a cosmology-side gate driven by DESI DR2 $w_0 w_a$ constraints, which fixes the required sign and amplitude, and an independent galaxy-side measurement of $a_0(z)$ from the deep-MOND baryonic Tully–Fisher (BTFR) zero point. We state the limitations plainly: the cosmology gate inherits $w(z)$ and is a consistency condition rather than independent evidence; the reference ratio at $z = 3$ is a CPL extrapolation beyond the supernova range; the predicted $a_0(z)$ is non-monotonic (a few-percent rise at $z \lesssim 0.5$ before declining at high z), so the two scales probe opposite branches of the same curve; and the galaxy-side test is at present M_{bar} -dominated and underpowered (neither passed nor failed). The central methodological finding redirects the observational strategy: in the BTFR estimator a_0 is 1:1 degenerate with M_{bar} , the dominant M_{bar} systematic (IMF, α_{CO} , dust) is *common-mode* and does not average down with sample size — with a $\gtrsim 0.08$ -dex correlated floor the BTFR test never reaches 3σ at any rotator count. The decisive lever is common-mode M_{bar} calibration to $\lesssim 0.05$ dex (with ~ 20 – 40 clean deep-MOND rotators), and the degeneracy-breaking observable is the rotation-curve/radial-acceleration-relation transition, not the BTFR zero point. At present DESI DR2 gives $R(z=3) = 0.775 [0.68, 0.88]$, a 2.0σ hint below our committed 3σ bar — undecided. We therefore pre-register — with a hashed, timestamped estimator, decision thresholds, and data hierarchy fixed before the data exist — the test against the forthcoming Rubin/LSST calibrated supernova cosmology sample (expected $\gtrsim 2027$), which forecasts project to settle the gate at the 3 – 6σ level by ~ 2028 – 2030 . The value of a_0 , the normalization Z , and the sign of the effect are posited, not derived; the kernel $\nu = \sqrt{1 + 1/y}$ is Milgrom’s (1999) and the covariant MOND-scale realization is AeST (Skordis & Zlosnik 2021). The contribution is a bounded, falsifiable, pre-registered program, not a detection.

Keywords: modified gravity, dark energy equation of state, supernova type Ia — standard candles, redshift surveys, rotation curves of galaxies.

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1 Introduction

1.1 The coincidence, and what promoting it to a law buys

A spiral galaxy’s mass discrepancy switches on below a universal acceleration $a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$ (Milgrom 1983 [1]). To order unity this number equals the acceleration scale set by the dark energy accelerating the whole universe, $c\sqrt{\rho_\Lambda}$ — a coincidence that is *not* original to this work: Limbach, Psaltis & Özel (2008) [3] identified it and tested candidate a_0 - H_0 and

a_0 - ρ_{DE} couplings against Tully–Fisher data to $z \simeq 1.2$. If the coincidence is physical — if a_0 is genuinely sourced by the dark sector — then as the dark-energy density changes through cosmic history, a_0 must change with it, on a track fixed by the cosmology with nothing to tune. This paper states that track precisely, shows what current data do to it, and pre-registers the decision procedure by which forthcoming data will judge it.

1.2 The framework and the horizon scale

Modified inertia (MI) proposes that the inertial response of a body weakens below a critical acceleration a_0 , rather than the gravitational field being modified. In the de Sitter–Unruh realization, an accelerated observer in a universe with a cosmological horizon sees a floor in the vacuum’s response set by the horizon temperature, and the transition scale is tied to the horizon:

$$a_0 = \frac{cH_\Lambda}{Z} = \frac{c^2\sqrt{\Lambda/3}}{Z}, \quad Z = \sqrt{\frac{32\pi}{3}} = 5.78881, \quad (1)$$

with $H_\Lambda \equiv c\sqrt{\Lambda/3}$ the de Sitter (pure dark-energy) Hubble rate. The Planck 2018 value $\Lambda = 1.089 \times 10^{-52} \text{ m}^{-2}$ gives the **canonical footing** $a_0^{\text{canon}} = c^2\sqrt{\Lambda/32\pi} = 9.355 \times 10^{-11} \text{ m s}^{-2}$ (pure ρ_{DE}). A second **ALT footing** replaces the pure- Λ density by the total present-day critical density, $a_0^{\text{ALT}} = cH_0/Z = 1.1305 \times 10^{-10} \text{ m s}^{-2}$. The two differ by 20.9%; both are carried on every dimensional number below (they cancel identically in every ratio).

Scope, stated once and binding throughout. Nothing here derives the *value* of a_0 , the constant Z , or the *sign* of the inertial correction: those are posited inputs. The low-acceleration interpolation used to read a_0 from rotation curves, $\nu(y) = \sqrt{1 + 1/y}$ with $y = g_{\text{bar}}/a_0$, is the identical functional form to Milgrom’s 1999 interpolation ([2], Eq. 9, in the original MOND context of [1]); Milgrom’s coefficient was $2cH_\Lambda$ rather than cH_Λ/Z . The covariant modified-*gravity* realization of a MOND scale is AeST [4]; this work adopts both and claims neither. The distinctive content claimed here is only the cH_Λ/Z *coefficient*, the modified-inertia reading, and the falsifiable $a_0(z) \propto \sqrt{\rho_{\text{DE}}(z)}$ law under test.

1.3 Related work by the author

This manuscript consolidates, into one peer-review-facing article, three of the author’s Zenodo preprint/pre-registration deposits released for reproducibility and timestamping: the parameter-free $a_0(z)$ law and its DESI confrontation [19], the non-circular cross-scale test [20], and the Rubin/LSST pre-registration [21]. Two companion papers by the author treat logically distinct results and are cited, not consolidated: a field-theory paper (covariant modified-inertia construction and an obstruction theorem; Zenodo record [22], PRD submission in preparation) and a galaxy-dynamics data paper (the SPARC a_0 -line slope measurement and its Λ -inversion; Zenodo record [23], MNRAS submission in preparation). On the footing fork the division of labor is explicit: *the value-normalization fork (canonical cH_Λ/Z vs cH_0/Z) is examined empirically in the companion SPARC measurement [23]; here only the evolution-law footing is under test*. Nothing in the present test depends on the covariant field-theory completion (Section 8).

2 The prediction

2.1 The parameter-free replacement of the free- w fit

The framework keeps General Relativity for the cosmological background and therefore has no native supernova formula: light-curve-to-distance standardization is unchanged, and the

Friedmann equation is the standard one. Its single distinctive move is that the dark-energy term is not a free fit but the galaxy acceleration scale,

$$H^2(z) = \frac{8\pi G}{3} \rho_m(z) + \frac{Z^2 a_0^2(z)}{c^2}, \quad \frac{Z^2 a_0^2}{c^2} = H_\Lambda^2 = \frac{\Lambda c^2}{3} \quad (z=0), \quad (2)$$

so that, inverting $Z^2 a_0^2/c^2 = \frac{8\pi G}{3} \rho_{\text{DE}}$ with $Z^2 = 32\pi/3$, the dark-energy mass density is $\rho_{\text{DE}}(z) = 4a_0^2(z)/(Gc^2)$ and

$$\boxed{a_0(z) = \frac{c}{2} \sqrt{G \rho_{\text{DE}}(z)} = c^2 \sqrt{\Lambda_{\text{eff}}(z)/32\pi} \implies R(z) \equiv \frac{a_0(z)}{a_0(0)} = \sqrt{\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE},0}}}} \quad (3)$$

The SI form $\frac{c}{2} \sqrt{G \rho_{\text{DE}}}$ uses the dark-energy *mass* density (kg m^{-3}); the equivalent geometric form uses $\Lambda_{\text{eff}} \equiv 8\pi G \rho_{\text{DE}}/c^2$ (m^{-2}). Both give $a_0^{\text{canon}} = 9.355 \times 10^{-11} \text{ m s}^{-2}$ at $z=0$ on Planck $\rho_{\text{DE},0}$; the two conventions must not be mixed. The redshift-tracking ratio $R(z)$ is convention-independent.

On the supernova side this adds **no free dark-energy parameter** — against ΛCDM 's Ω_Λ (and, in the evolving-DE extension, w_0, w_a). The framework does not, however, produce a parameter-free dark-energy history from nothing: it *consumes* whatever $\rho_{\text{DE}}(z)$ the cosmic data prefer (itself the w_0, w_a fit to distances) and requires that the *same* function reproduce the galaxy-side $a_0(z)$. The content is therefore a cross-*dataset* constraint, not a first-principles prediction of $w(z)$. At $z=0$ the framework is identical to ΛCDM by construction. It differs from ΛCDM only if $a_0(z)$ evolves.

2.2 R is Z -independent, and the two distinct footing forks

Because R is a ratio of a_0 at two redshifts, the normalization Z and the absolute value of a_0 cancel identically: both *value* footings (canonical 9.36×10^{-11} and ALT $1.13 \times 10^{-10} \text{ m s}^{-2}$) collapse onto the same $R(w_0, w_a)$. R depends on the dark-energy expansion history alone. One clarification is owed, because the framework's internal audit distinguishes two separate footing forks. The fork that cancels is the *value* fork (which absolute a_0 today). A second, logically independent fork concerns the *evolution law* itself: the adiabatic-horizon $a_0 \propto \sqrt{\rho_{\text{DE}}(z)}$ (which declines at high z under evolving DE) versus a rival $a_0 \propto cH(z) = cH_0 E(z)$ (which *rises* with z). **This work commits to the $\sqrt{\rho_{\text{DE}}}$ footing as the prediction under test**, in advance: if the data instead showed a rise, that registers as Verdict C (framework strained) in the frozen protocol of Section 6 — it is not re-interpreted post hoc as the rival footing succeeding.

2.3 What the framework predicts, and what it inherits

For the CPL parameterization $w(a) = w_0 + w_a(1-a)$,

$$\frac{\rho_{\text{DE}}(z)}{\rho_{\text{DE},0}} = (1+z)^{3(1+w_0+w_a)} \exp\left(\frac{-3w_a z}{1+z}\right). \quad (4)$$

Two reference cosmic tracks bracket the question, with the rival evolution model shown for contrast:

track	(w_0, w_a)	$a_0(3)/a_0(0)$	$a_0(3.25)/a_0(0)$
flat- Λ ($w = -1$)	$(-1.00, 0.00)$	1.000 (flat)	1.000
DESI-DR2 evolving	$(-0.75, -0.90)$	0.712 (declining)	0.685
(rival $cH(z)$ evolution model)	$a_0 \propto cH_0 E(z)$	4.57 (rising)	4.99

The framework’s distinctive canonical-footing prediction is a **mild decline**, $a_0(3)/a_0(0) \approx 0.60\text{--}0.75$, if the DESI-preferred evolving dark energy is real. If instead $w \rightarrow -1$, the framework reduces to Λ CDM at all z and the distinctive signal vanishes, leaving only the $z = 0$ tie. The third row is *not* the ALT $z = 0$ normalization but a distinct evolution model in which a_0 tracks the total-density Hubble rate and therefore rises steeply — the opposite sign. The flat-DESI separation is only 0.147 dex at $z = 3$ (0.164 dex at $z = 3.25$); that small gap is what the galaxy data must resolve.

To be precise about the falsifiable content (and to avoid an overstated biconditional): R tracks $\sqrt{\rho_{\text{DE}}(z)}$, whatever that history is. Under the DESI-preferred ($w_0 > -1$, $w_a < 0$) CPL histories R declines at high redshift; it reduces to $R = 1$ exactly if $w = -1$; and histories with ρ_{DE} *rising* toward high z (e.g. a constant $w > -1$) would produce a rising R , which under the frozen protocol registers as Verdict C (strained) when read at the frozen reference redshift (Section 6).

The adiabatic-horizon ansatz, stated as an ansatz. The horizon relation (1) is derived for a *static* de Sitter horizon (constant Λ). Promoting it to $a_0(z) = \frac{c}{2} \sqrt{G\rho_{\text{DE}}(z)}$ for an evolving equation of state is an **adiabatic / instantaneous-horizon ansatz** — the assumption that the inertia scale tracks the instantaneous dark-energy density — not a result derived from the de Sitter–Unruh construction, whose horizon is that of a constant- Λ background. The z -tracking under test is therefore a test of this ansatz combined with the measured $w(z)$; a null would disfavour the ansatz, not the static-horizon relation it reduces to at $z = 0$. Relative to [3], who treated $a_0 - H_0$ and $a_0 - \rho_{\text{DE}}$ as two candidate couplings, the framework’s addition is to *fix* the coupling to ρ_{DE} via this ansatz (itself posited) — so the choice between H_0 and ρ_{DE} is made by the construction, not left free — and to commit a hashed decision procedure against the Rubin era.

Non-monotonicity. Under an evolving CPL history $a_0(z)$ is *not monotonic*: at the DESI-central rate ($w_0 = -0.838$, $w_a = -0.62$) it rises to a small bump — R peaks at ≈ 1.036 near $z \approx 0.35$ — and only crosses back below unity at $z \approx 0.9$, declining at higher redshift. (On the DESI+CMB+DESY5 central history, $w_0 = -0.752$, $w_a = -0.86$, the bump is +6% peaking at $z \approx 0.41$, the phantom-divide crossing [19].) The consequences of summarizing this curve by a single high- z point are drawn out in Section 8.

The sharpest galaxy-side signature is a sign. In the deep-MOND regime $V^4 = GM_{\text{bar}}a_0$, so $V \propto a_0^{1/4}$ and $d \log V = \frac{1}{8} d \log \rho_{\text{DE}}$. On the DESY5-central history ($R(3) = 0.737$) the framework predicts high- z deep-MOND discs lying 0.033 dex (−7.3% in V) *below* the local BTFR at $z = 3$, where standard fixed- a_0 MOND sits on it and the rival rising model sits +0.166 dex above [19]. The sign is more robust to amplitude systematics than the a_0 value itself.

3 The non-circular cross-scale strategy

Λ CDM gives no reason for a galaxy’s internal acceleration scale to track cosmic expansion; the framework forces exactly that link. Crucially, the equality of a galaxy a_0 and a cosmic Λ at a *single* epoch is a definition — one can always convert one into the other, and reading agreement there is circular. The genuine, non-circular test is whether **two independently measured functions of redshift** — a_0 from galaxy kinematics (parsec–kiloparsec scales) and ρ_{DE} from cosmic distances (gigaparsec scales) — track each other as z varies. The framework *inherits* the dark-energy history $w(z)$; its content is the tie between the two datasets, so any decline

it exhibits is a consequence of the framework relation combined with the measured $w(z)$, not an independent statement about dark energy.

3.1 Definition versus test, and the $z = 0$ anchor

The $z = 0$ equality $a_0(0) = (c/2)\sqrt{G\rho_{\text{DE},0}}$ is definitional: a $z = 0$ match is a consistency knot, not an independent success. Its value is as the anchor the bridge extends from. The Λ -blind galaxy $a_0(0)$ is taken from the gas-dominated SPARC BTFR zero point measured without any cosmological- Λ input (companion measurement [23]): $a_0(0) = 1.181 \times 10^{-10} \text{ m s}^{-2}$ ($\pm 16\%$). Against the cosmic-side $(c/2)\sqrt{G\rho_{\text{DE},0}}$:

footing	cosmic $a_0(0)$	$a_{0,\text{SPARC}}/a_{0,\text{cosmic}}$	tension
Planck ($\Omega_{\text{DE}} = 0.685$, $H_0 = 67.4$)	9.36×10^{-11}	1.26	1.44σ
SHOES- H_0 ($\Omega_{\text{DE}} = 0.685$, $H_0 = 73.0$)	1.01×10^{-10}	1.17	0.95σ

The tie holds at $\sim 1\sigma$ on the canonical (Planck- H_0) footing and slightly better at the local H_0 . This is by construction and carries no evidential weight beyond internal consistency; the non-circular content is entirely in the *slope*.

3.2 Why a wrong background cannot manufacture the signal

Because the two datasets never share inputs, a wrong background assumption shifts both together and cannot manufacture a spurious co-evolution. The mild distance–cosmology dependence of the galaxy side is quantified: $M_{\text{bar}} \propto D^2$, and the ratio $D(\text{DESI})/D(\text{flat})$ shifts M_{bar} by ≤ 0.015 dex at $z = 1$ (smaller at higher z) — roughly $10\times$ below the signal and $15\times$ below the single-object error of Section 4.2. Testing whether a_0 tracks ρ_{DE} is not the same as assuming the DE evolution.

4 The galaxy-side $a_0(z)$ from the high- z BTFR

4.1 The estimator

On the galaxy side, $a_0(z)$ is read from the BTFR zero point at each redshift. In the deep-MOND regime $V^4 = a_0 G M_{\text{bar}}$ is exact (the fitted slope is forced to 4), so at fixed circular velocity

$$\frac{a_0(z)}{a_0(0)} = \frac{(V_z/V_0)^4}{M_{\text{bar},z}/M_{\text{bar},0}}, \quad \text{equivalently} \quad \Delta \log a_0 = -\Delta \log M_{\text{bar}}|_{\text{fixed } V}. \quad (5)$$

Reading $a_0(z)$ is the same operation as reading the BTFR zero point. The estimator is valid *only* where the fitted slope is genuinely 4, i.e. for objects deep in the low-acceleration regime $g \ll a_0$.

4.2 The one high- z datum: the Big Wheel at $z = 3.25$, now transitional

The best available high- z candidate is the “Big Wheel” (Wang et al. 2025 [14]; $z = 3.2514$), a large low-acceleration disk. A Monte Carlo over the discovery-paper velocity and baryonic-mass systematics ($V_{\text{rot}} = 280 \pm 31 \text{ km s}^{-1}$ with asymmetric-drift correction; $M_\star = 1.7 \pm 0.7 \times 10^{11} M_\odot$ dynamically capped; $M_{\text{gas}} = 1.8 \pm 0.9 \times 10^{11} M_\odot$, $\times 1.36$ He) gave $a_{0,\text{eff}} = 1.54 (+1.10/-0.61) \times 10^{-10} \text{ m s}^{-2}$, i.e. ratio $1.31 (+0.93/-0.52)$ against the SPARC anchor (1.65 against canonical, 1.37 against ALT — both footings carried). On those masses the

datum was consistent with a constant and with the DESI decline, and disfavored the rising $cH(z)$ evolution model ($\sim 5\times$) at $\sim 2\sigma$.

That discovery-mass state is superseded. A dedicated 2026 ALMA dynamical model (Quadri et al. 2026 [15]; W. Wang among the authors) finds a mildly rising rotation curve reaching $V_{\text{rot}} \approx 314 \text{ km s}^{-1}$ at the outermost *measured* radius $\approx 16.5 \text{ kpc}$, $M_{\text{gas}} = 10^{10.76} M_{\odot}$, and — critically — a stellar mass ambiguous by 0.37 dex: the dynamical-model fiducial $M_{\star} = 10^{11.00} M_{\odot}$ versus a higher SED/half-mass value $10^{11.37} M_{\odot}$ that the authors flag as in tension with the dynamics. Applying the framework’s exact kernel $a_0 = (g_{\text{obs}}^2 - g_{\text{bar}}^2)/g_{\text{bar}}$ at the measured 16.5-kpc point, the inferred a_0 swings from ratio 0.86 (SED $M_{\star}$) to 3.25 (dynamical $M_{\star}$) at central values — Monte-Carlo medians 1.28 [0.41, 2.60] and 3.11 [1.89, 4.71] respectively, with $\sim 28\%$ of the SED- $M_{\star}$ posterior unphysical ($g_{\text{bar}} > g_{\text{obs}}$). The object sits at $y = g_{\text{bar}}/a_0 \approx 0.7\text{--}1.3$ — *transitional, not deep-MOND* — so a_0 is hyper-sensitive to M_{bar} , and the 0.37-dex stellar-mass ambiguity alone spans a factor ~ 3.8 in a_0 (0.4–4.7 across full systematics). Three consequences, stated plainly: (i) the single Big Wheel a_0 remains M_{bar} -hostage and cannot separate the framework’s ~ 0.68 decline at $z = 3.25$ from a strong rise; (ii) the $\sim 2\sigma$ exclusion of the rising model held *only on the discovery masses* — under the authors’ fiducial dynamical mass the object itself sits at a $\sim 3\times$ rise (3.1 [1.9, 4.7]), only $\sim 1.2\sigma$ from the $\sim 5\times$ rising track, so the rise-exclusion too is M_{bar} -hostage; and (iii) this is an honest downgrade of the one in-hand galaxy-side value, and it *empirically confirms* the central finding of Section 5 on the very object one would reach for: with new, dedicated dynamical data the single-object a_0 did not tighten, because M_{bar} calibration, not the object, is decisive. (Scripts: `galaxy_a0z.py` for the discovery-mass Monte Carlo, `bigwheel_update.py` for the v2 re-extraction; Appendix A.)

4.3 Why intermediate- z is not a clean a_0 probe

The many $z \sim 0.5\text{--}2.3$ rotating disks (e.g. KMOS^{3D}) do not help, for a physical reason: these massive rotators sit at $g \gtrsim a_0$, *not* deep-MOND, so their fitted BTFR slopes are 3.0–3.85, not 4, and the zero point is degenerate with the slope, the pivot, and baryon/dark-matter-fraction evolution. The inferred “ a_0 evolution” flips sign with the analysis choice: Übler et al. (2017) [11] (fixed local slope 3.75) reads a_0 rising $\sim 2\text{--}3\times$; Sharma et al. (2024) [13] (free slope 3.21) the opposite sign; Di Teodoro et al. (2016) [10] and Tiley et al. (2019) [12] read null. The scatter across analyses (a factor $\sim 2\text{--}3$, in either direction) dwarfs the 0.70-vs-1.0 signal. These points are not valid a_0 readouts; where the $z \approx 2.3$ point enters the joint fit below it is placed at its fixed-slope value 1.86 (rising — the *wrong* sign for the framework decline), so it counts against, not for, the framework.

4.4 The a_0 – M_{bar} degeneracy

Because $\Delta \log a_0 = -\Delta \log M_{\text{bar}}$ at fixed V , a mis-estimate of the baryonic mass propagates 1:1 into a_0 : the two are algebraically degenerate for this estimator. Rising molecular-gas fractions ($\sim 10\%$ at $z = 0$ to $\sim 50\%$ at $z = 2$), the CO-to- H_2 conversion α_{CO} , dust, and IMF evolution all move the BTFR zero point and are indistinguishable from an a_0 shift with the BTFR / point-mass method. This is the crux limitation developed in Section 5.

4.5 Does the galaxy data show the decline? No — neither way

Assembling the honest galaxy points (`confront.py`, Appendix A):

set	$\chi^2(\text{flat})$	$\chi^2(\text{DESI-decl})$	$\Delta\chi^2$	lean
all 4 points	1.08	3.05	-1.97	flat, weak
$z = 0$ + Big Wheel (discovery masses; transitional per Sec. 4.2)	0.27	1.55	-1.28	flat, weak

The second row is the sole formally meaningful statistic; the “all 4 points” row is illustrative only, because the intermediate- z points are declared invalid a_0 readouts (Section 4.3) and cannot legitimately enter a likelihood. Both rows agree in direction: the data mildly prefer flat, $|\Delta\chi^2| \approx 1.3\text{--}2.0$ — not significant — and what lean exists in the intermediate- z points is toward *rising*, the wrong sign for the framework decline. The distinctive 0.60–0.75 decline is neither detected nor excluded. The separability statistic — the maximum sigma at which the current errors could ever tell flat from the DESI decline — is $S = 0.73\sigma$ (Big Wheel alone), 0.73σ (clean pair), and 0.80σ (all four points): **underpowered, not passed, not failed**. Every galaxy point sits $\leq 1.2\sigma$ from both tracks.

5 The key finding: M_{bar} calibration, not rotator count, is decisive

The decisive question for any future test is how the significance scales. The naive forecast assumes independent per-object errors that beat down as $1/\sqrt{N}$: the target signal is $-\log_{10}(0.70) = 0.155$ dex, requiring $\sigma_{\text{mean}} = 0.052$ dex for 3σ , hence $N \approx 4$ rotators at SPARC-like precision (0.10 dex) or $N \approx 24$ at realistic high- z per-object precision (0.25 dex). **This is wrong**, because the dominant M_{bar} error — α_{CO} , IMF, and dust prescriptions applied *identically* to a sample — is largely common-mode, and a common-mode error does not average down with N . With $\sigma_{\text{mean}}^2 = \sigma_{\text{ind}}^2/N + \sigma_{\text{cm}}^2$ and a correlated M_{bar} floor σ_{cm} :

M_{bar} common-mode floor	N for 3σ
0.00 dex	24
0.05 dex	~ 377
≥ 0.08 dex	never reaches 3σ at any N

Because realistic high- z M_{bar} calibration floors are $\sim 0.08\text{--}0.15$ dex (α_{CO} alone is a 0.1–0.2 dex systematic), the test is decisive **only if the common-mode M_{bar} systematic can be driven below ~ 0.05 dex** — a calibration problem, not a counting problem. The 2026 ALMA re-analysis of the Big Wheel (Section 4.2) demonstrates this on real data: a 0.37-dex stellar-mass ambiguity on a single dedicated-data object swings its a_0 by a factor ~ 3.8 — exactly the M_{bar} -hostage behaviour this section predicts, and precisely why more objects at fixed M_{bar} calibration would not help. This is the paper’s core methodological contribution: **the decisive observational lever is a uniformly calibrated M_{bar} ladder to $\lesssim 0.05$ dex common-mode, not the number of rotators**. (Both footings enter identically — the forecast is in ratios and dex, so the 20.9% footing choice cancels.)

5.1 Forecast requirements, correctly stated

Reaching 3σ on the 0.60–0.75 decline versus flat therefore requires *both* (i) a common-mode M_{bar} calibration $\lesssim 0.05$ dex *and* (ii) a sample of $\sim 20\text{--}40$ clean deep-MOND $z \sim 2\text{--}3$ rotators, Big-Wheel-like, with gas-traced outer rotation curves (JWST/ALMA) or individually resolved deep-MOND curves (ELT/HARMONI); the per-object precision required is ≤ 0.16 dex ($N = 10$), ≤ 0.23 dex ($N = 20$), ≤ 0.33 dex ($N = 40$), all under the independence assumption

that returns the argument to the common-mode floor above. Neither condition alone suffices: without the calibration floor, no rotator count reaches 3σ . Clean deep-MOND disks like the Big Wheel are rare, so assembling 20–40 is itself an observational stretch.

5.2 The a_0 -separable observable

The deeper methodological escape is to stop using the BTFR zero point at all. Because a_0 and M_{bar} are 1:1 degenerate in the BTFR, no amount of BTFR data isolates a_0 from baryonic-mass evolution. The a_0 -**separable** observable is the *shape* of the rotation curve — the radial-acceleration-relation (RAR) transition — which depends on a_0 at fixed M_{bar} and therefore breaks the degeneracy the zero point cannot. A high- z RAR-transition measurement reads a_0 directly, without hostage to the gas-mass normalization. This is the observable to target: not more BTFR zero points, but resolved high- z rotation-curve shapes.

6 The cosmology-side gate and the pre-registration

6.1 DESI DR2 propagated through the relation

On the cosmic side there is no free dark-energy parameter to fit — the framework consumes whatever $\rho_{\text{DE}}(z)$ the data prefer. Propagating the DESI DR2 [5] w_0w_a CDM posterior (BAO + CMB + SNe; representative marginals with w_0 – w_a correlation ≈ -0.87 ; Λ CDM excluded at 2.8σ /Pantheon+, 4.2σ /DESY5, 3.8σ /Union3) through Eq. (3) (`desi_posterior_a0z.py`, Appendix A):

SNe combination	$a_0(3)/a_0(0)$	[16%, 84%]	$P(\text{declines})$	signif.	in 0.60–0.75
DESI+CMB+Pantheon+	0.775	[0.683, 0.878]	97.9%	2.0σ	38%
DESI+CMB+DESY5	0.737	[0.651, 0.833]	99.3%	2.5σ	51%
DESI+CMB+Union3	0.707	[0.599, 0.831]	98.3%	2.1σ	48%

Across the three combinations the median is $a_0(3)/a_0(0) \simeq 0.74$ with a $\sim 2.2\sigma$ decline, and all three posteriors overlap the framework’s 0.60–0.75 band. This is a genuine, live cosmology-side hint: *if* the DESI evolving-dark-energy signal is real, the induced $a_0(z)$ decline falls inside the framework band. It is not, however, an independent test — it is a re-expression of the DESI $w(z)$ result through the framework relation, the same DESI information seen twice, and it cannot be added to the galaxy-side χ^2 as if it were a second measurement. It dissolves entirely if the DESI signal regresses to $w = -1$.

6.2 The frozen estimator and pre-committed thresholds

The decision procedure was frozen, SHA-256-hashed, and Zenodo-timestamped on 2026-07-21 [21], before any calibrated Rubin supernova cosmology exists (LSST began operations 2026-06-30; the LSST-DESC calibrated Type-Ia sample is not expected before ~ 2027). The estimator (`a0z_gate_estimator.py`) draws $N = 4 \times 10^5$ samples from the input (w_0, w_a) bivariate Gaussian, maps each through Eq. (3) to $R = a_0(3)/a_0(0)$, and reports the median, [16, 84] interval, and the posterior credibility of a decline mapped to a Gaussian sigma. **No threshold, band, or estimator may change after the freeze; only the input (w_0, w_a) is updated.**

Verdict	Pre-committed condition
A) GATE OPEN — distinctive prediction alive and supported	decline at $\geq 3\sigma$ and $R \in [0.55, 0.85] \rightarrow a_0$ evolves as required; warrants the independent galaxy-side confrontation. GATE OPEN means “survives to face the galaxy-side test,” explicitly not “confirmed.”
B) GATE DIS-SOLVED — safe core, <i>not</i> falsification	consistent with flat ($R = 1$ within 2σ) $\rightarrow$ framework reduces to constant- a_0 MOND; the distinctive prediction is inapplicable.
C) FRAMEWORK STRAINED — cosmology-side tension	$R > 1$ (a rise) at $\geq 3\sigma \rightarrow$ the scale rises with z , which $a_0 \propto \sqrt{\rho_{\text{DE}}}$ does not permit under evolving DE.
UNDECIDED	none of the above met.

The band $[0.55, 0.85]$ is fixed and hashed. Its width comes from the spread of currently-allowed dark-energy histories — holding $w_a = -0.62$: $w_0 \approx -0.79 \rightarrow R = 0.85$ (mild evolution), $w_0 \approx -1.00 \rightarrow R = 0.55$ (strong evolution) — *not* from the two value footings (which give an identical R at fixed (w_0, w_a) and cannot widen it). Two honesty notes: (i) the band is deliberately wider than the nominal distinctive claim $R \approx 0.60\text{--}0.75$, and the DESI value 0.775 sits just above the nominal 0.75 though inside the operational band; (ii) the upper edge 0.85 is close to flat, so “supported” at that edge is a weak decline — the intended cost of a band broad enough not to be tuned to one posterior. A decline steeper than $R < 0.55$ at $\geq 3\sigma$ returns UNDECIDED, not a verdict — the frozen procedure has no “over-decline strain” branch mirroring Verdict C.

6.3 Data hierarchy and the refused DIY-Hubble-diagram path

The gate is fed by a strict, frozen hierarchy: (1) **primary** — the LSST-DESC *calibrated* Type-Ia sample ($\sim 2027+$), the only input that can produce a supported verdict; (2) **interim** — published Rubin $w_0 w_a$ constraints as they appear; (3) **readiness tracker only** — a running count of broker-classified Type-Ia candidates from the public alert stream (`broker_sn_counter.py`), which *never* feeds the verdict. We explicitly refuse to build a do-it-yourself Hubble diagram from public broker light curves: without the DESC calibration layer (zero-point stability, photometric-redshift systematics, modeled selection function), any $w(z)$ extracted from raw broker photometry is systematics-dominated, and a “Rubin supports $a_0(z)$ ” claim built on it would be a manufactured result.

6.4 State at the freeze (2026-07-21)

Running the frozen estimator on the best current input — DESI DR2 [5], $w_0 = -0.838 \pm 0.055$, $w_a = -0.62 \pm 0.22$, correlation -0.86 (the DESI+CMB+Pantheon+ combination, the mildest; DESY5 and Union3 prefer stronger evolution and the verdict is unchanged under them) — gives $R = 0.775 [0.683, 0.878]$, a decline at 2.0σ : below the committed 3σ threshold, hence **UNDECIDED** — a hint, explicitly not a detection, explicitly not gate-open. The broker readiness tracker stood at 36,836 Type-Ia-classified objects at the freeze; by the frozen rule it feeds timing only, never the verdict.

7 Forecast against Rubin/LSST

The committed forecast (`forecast_rubin_a0z.py`) propagates representative projected Rubin $w_0\text{--}w_a$ covariances (photometric SN figure-of-merit ~ 150 alone, ~ 500 with CMB/BAO

[18, 17]) through Eq. (3), taking the DESI-central evolving history as the assumed truth:

Input scenario	$\sigma(w_0)$	$\sigma(w_a)$	$R = a_0(3)/a_0(0)$	Decline signif.
DESI DR2 now (baseline)	0.055	0.22	0.775 [0.683, 0.878]	2.0σ
Rubin SN alone (FoM ~ 150)	0.040	0.15	0.775 [0.717, 0.837]	3.3σ
Rubin SN + CMB/BAO (FoM ~ 500)	0.020	0.08	0.775 [0.742, 0.809]	$4.6\text{--}4.7\sigma$

The **mirror test**: if the universe is in fact flat ($w = -1$, a_0 truly constant), Rubin-alone pins $R = 1.00 \pm 0.078$, excluding 0.74 at 3.3σ ; Rubin + CMB/BAO gives $R = 1.00 \pm 0.043$, excluding it at 6.0σ . The honest summary is deliberately unheroic: if dark energy evolves at the DESI-central rate, Rubin sharpens the decline from $\sim 2.0\sigma$ to $\sim 3.3\sigma$ (SNe alone) or $\sim 4.6\text{--}4.7\sigma$ (with CMB/BAO); if dark energy is flat, Rubin excludes the framework’s 0.74 at $\sim 3.3\text{--}6.0\sigma$. Either way the gate is likely settled at the $3\text{--}6\sigma$ level by roughly 2028–2030 — a real upgrade on DESI-now, not a one-instrument slam dunk. These are projections, not data, and cannot produce a real verdict.

8 Limitations

Because the estimator and thresholds are hashed and cannot be revised after the freeze, the following limitations are documented rather than fixed. They are stated here so that no reading can present the test as tighter than it is.

1. **The signal is a $z = 3$ extrapolation beyond the supernova range, and the significance is z -dependent.** The LSST photometric SN sample constrains $w(z)$ mainly at $z \lesssim 1.2$; the gate is read at $z = 3$. Propagating the *same* DESI-DR2 posterior to different reference redshifts gives a decline significance of 0.34σ at $z = 1$ ($R = 0.99$, effectively flat), 1.7σ at $z = 2$, 2.0σ at $z = 3$, and 2.4σ at $z = 10$; for the tight SN+CMB/BAO forecast covariance it is 1.1σ at $z = 1$ but 4.7σ at $z \geq 2$ (reproduced by `significance_vs_z.py`, Appendix A). What the gate genuinely tests is whether the SN-constrained posterior excludes $(w_0, w_a) = (-1, 0)$; the $z = 3$ evaluation converts that into an a_0 -ratio but adds no independent $z = 3$ information. $R(z=3) = 0.775$ must *not* be read as a measured $z = 3$ acceleration scale.
2. **The framework’s own $a_0(z)$ bumps up at low redshift, which is where the galaxy-side test lives.** At the DESI-central history R peaks at ≈ 1.036 near $z \approx 0.35$ and only falls below 1 beyond $z \approx 0.9$. A GATE-OPEN verdict on the $z = 3$ decline does not translate into a low- z decline for the galaxy side to corroborate; at $z \lesssim 0.5$ the framework predicts a slight *rise* (a few percent), at $z \sim 1$ essentially no change. Anyone carrying “the framework predicts a_0 declines” to low- z observers would have the sign wrong. The two lanes probe opposite branches of the same non-monotonic curve.
3. **Verdict C is well-posed only at the frozen $z = 3$.** Because $a_0(z)$ genuinely rises for $z \lesssim 0.9$, a measured rise contradicts the $\sqrt{\rho_{\text{DE}}}$ footing only when read at high z ; Verdict C must never be applied to a low- z input.
4. **Doc-vs-code items in the frozen protocol.** (a) The Verdict-B sub-clause (“if $R \leq 0.85$ is additionally excluded at $\geq 3\sigma$, the decline is positively ruled out”) is narrative only; the frozen script computes Verdict B purely as consistency with $R = 1$ within 2σ . (b) The over-decline gap ($R < 0.55$ significant $\rightarrow$ UNDECIDED) is a real asymmetry with Verdict C. (c) The frozen estimator prints GATE OPEN for the two

forecast covariances; those are fabricated inputs shown for illustration and, by the data hierarchy, can never constitute a verdict.

5. **Significance convention.** The reported “ $N\sigma$ ” is a Bayesian posterior tail (the fraction of samples with $R \geq 1$ mapped to a Gaussian sigma), not a frequentist likelihood-ratio detection; the estimator ($N = 4 \times 10^5$) and forecast script ($N = 3 \times 10^5$) differ at the 0.1σ level (4.7 vs 4.6σ), reported as the range.
6. M_{bar} **dominance of the galaxy side.** The independent lane is currently underpowered ($S \simeq 0.8\sigma$), M_{bar} -hostage (Sections 4.2–5), and its intermediate- z literature is sign-contested.
7. **The covariant completion is not in hand, and the test does not lean on it.** The observable R requires only the scalar relation Eq. (3) and a $w(z)$ posterior — it does not invoke a Lagrangian. A separate modified-inertia field-theory program exists [22] but this test is scoped so that nothing rests on it; the absence of a completed covariant modified-inertia theory is a limitation of the framework, disclosed, not of the test.
8. **The adiabatic-horizon ansatz is posited** (Section 2.3), and the evolution-law fork (Section 2.2) is resolved by pre-commitment, not by derivation.

9 Conclusion

We have set out, and pre-registered, a non-circular cross-scale test of de Sitter–Unruh modified inertia: the framework replaces Λ CDM’s free dark-energy fit with the galaxy acceleration scale, forcing $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE},0}}$ — a bridge between galaxy kinematics and cosmic distances that Λ CDM structurally lacks. The $z = 0$ tie holds at $\sim 1\sigma$ (the definitional anchor). The cosmology side is live but inherited: the DESI DR2 posterior propagated through the relation yields $a_0(3)/a_0(0) \simeq 0.74$ at $\sim 2.2\sigma$ — below the committed 3σ bar, hence UNDECIDED — and it dissolves if $w \rightarrow -1$. The galaxy side is underpowered: the distinctive 0.60–0.75 decline is neither detected nor excluded ($S \simeq 0.8\sigma$); the one $z = 3.25$ object is transitional and M_{bar} -hostage across ratio 0.4–4.7, and even its former $\sim 2\sigma$ exclusion of the rising model held only on the discovery masses. The central finding is methodological and redirects the observational strategy: the dominant M_{bar} systematic is common-mode, so with a $\gtrsim 0.08$ -dex correlated floor the BTFR test never reaches 3σ at any rotator count; the decisive lever is M_{bar} calibration to $\lesssim 0.05$ dex common-mode, and the a_0 -separable observable is the rotation-curve/RAR transition, not the BTFR zero point. The estimator, thresholds, data hierarchy, and refusal of the DIY-Hubble path are hashed and timestamped before the deciding Rubin data exist; forecasts project a 3–6 σ settlement by ~ 2028 –2030. The value of a_0 , the normalization Z , and the sign of the effect are posited, not derived; both footings are carried throughout; and the contribution is a bounded, falsifiable, pre-registered program, not a detection.

Data and code availability

Every load-bearing quantitative claim in this manuscript traces to a committed, runnable, exit-0 Python script (numpy/scipy) in the author’s public repository, which is *self-verifying*: re-running the committed scripts reproduces every number in the text and tables (Appendix A lists the inventory and the freeze hashes). The verification toolkit is released as **a0kit** (Zenodo software concept DOI 10.5281/zenodo.21478981; an ASCL submission is pending editor review). The consolidated source deposits are Zenodo DOIs 10.5281/zenodo.20737162

(the $a_0(z)$ law and DESI confrontation [19]), 10.5281/zenodo.21440407 (the cross-scale test [20]), and 10.5281/zenodo.21478568 (the Rubin/LSST pre-registration [21]); companions are 10.5281/zenodo.21403470 (field theory [22]) and 10.5281/zenodo.21419735 (the SPARC a_0 -line measurement [23]).

A Freeze integrity and script inventory

Freeze hashes

The SHA-256 hashes of the frozen artifacts, recorded in `FREEZE_HASHES.txt` and verified 2026-07-21 [21]:

```
852ca954431b9239415038dde9f2b2ad9f89fe6325ff7a8ba3e3ebe357366740  a0z_gate_estimator.py
ba24927ddb3263c390646019b1dba1968c4aa3de5ed572bc5de57685afd2b77f  broker_sn_counter.py
bd714c93edd3eadc6c84413dee331e86be7ebaca8f0a5f88c84ec1542f46df0f  PREREGISTRATION.md
530e66beef425bb500d085051c3219669e6d6e31364eec9263ea036bca7a5697  forecast_rubin_a0z.py
```

No threshold, band, or estimator changes after these hashes; only the calibrated (w_0, w_a) input is updated when the LSST-DESC sample lands. The Zenodo deposit timestamp [21] is the public lock-in.

Script inventory

All numbers in this manuscript are reproduced by the following committed scripts (numpy/scipy only, exit 0, both footings carried):

- `a0z_gate_estimator.py` (frozen) — the gate: verdict logic, thresholds, R posterior; reproduces $R = 0.775 [0.683, 0.878]$, 2.0σ , UNDECIDED.
- `forecast_rubin_a0z.py` (frozen) — the Rubin forecast table and the mirror test.
- `broker_sn_counter.py` (frozen) — the readiness tracker (timing only; never feeds the verdict).
- `desi_posterior_a0z.py` — the three-SNe-combination DESI propagation table of Section 6.1.
- `galaxy_a0z.py` — the galaxy-side ladder: SPARC anchor, sign-contested intermediate- z compilation, discovery-mass Big-Wheel Monte Carlo.
- `bigwheel_update.py` — the v2 ALMA-dynamical-model re-extraction of the Big Wheel a_0 (both stellar-mass scenarios; the 0.4–4.7 ratio swing of Section 4.2).
- `confront.py` — the joint χ^2 , separability S , $z = 0$ tie, and the common-mode-floor forecast table of Section 5.
- `significance_vs_z.py` (post-freeze traceability addendum; deliberately *not* in the freeze hashes) — the z -dependent decline significances of Section 8, item 1.

Supporting reconstructions from an independent lane confirm that Type Ia supernovae alone cannot measure $a_0(z=3)$: the dark-energy residual $E^2 - \Omega_m(1+z)^3$ is a difference of large numbers, and the model-independent Gaussian-process reconstruction [16] is sign-pinned only to $z \approx 0.40$ — which is exactly why the two-dataset (galaxy $\times$ cosmic-distance) construction, and not SNe alone, is the test.

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